

1. Setup e Imports

```
# Instalar SAELens si no está disponible:
# !pip install sae-lens

import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import math
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../..'))
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkpoint_dir
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, get_logs_dir

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4, suppress=True)

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Dispositivo: {device}')
```

Dispositivo: cuda

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'topk',          # Arquitectura del SAE
    'tecnica': 'batch-topk',    # BatchTopK
    'capa': 6,                  # Layer del modelo OthelloGPT
    'juegos': 100000,           # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',
    'experiment_base_path': os.path.abspath('../..../experiments'),
    'extras': {
        'expansion': 8,        # d_sae / d_in = 8192 / 512
        'k': 50,               # número de features activas L0 promedio    k
        'epochs': 1000,
        'patience': 20
    }
}

print('\nMetadata de naming:')
for key, value in NAMING_META.items():
    print(f'  {key}: {value}')
```

Metadata de naming:

```
variante: topk
tecnica: batch-topk
capa: 6
juegos: 100000
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\models\experiments
extras: {'expansion': 8, 'k': 50, 'epochs': 1000, 'patience': 20}
```

```
extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
    NAMING_META['extras']
```

```

) if NAMING_META.get('extras') else None

# get_plots_dir crea el directorio automáticamente
plots_dir = Path(get_plots_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
))

print(f'extras_id: {extras_id}')
print(f'plots_dir: {plots_dir}')

```

Nuevo extras registrado: ex1 → {'expansion': 8, 'k': 50, 'epochs': 1000, 'patience': 20}
extras_id: ex1
plots_dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\models\experiments\layer_06\sae_topk_topk\100000games\ex1\plots

3. Configuración del SAE con SAELens

Configuración usando LanguageModelSAERunnerConfig de SAELens con `architecture='batchtopk'`.

Arquitectura BatchTopK: - **BatchTopK activation:** calcula un threshold global por batch para que el **promedio** de features activas por muestra sea **k** (L0 promedio **k**, pero varía por muestra) - **Sin coeficiente de sparsidad:** la sparsity se controla directamente con **k** - **Loss = MSE únicamente** (sin penalización L1)

Parámetro	L1 (standard)	TopK	BatchTopK
Sparsity	sparsity_coef (soft)	k (hard, por muestra)	k (promedio del batch)
L0	Variable	Fijo = k por muestra	Variable por muestra, promedio k
Loss	MSE + $\cdot$ L1	MSE	MSE

Nota sobre el runner: `LanguageModelSAETrainingRunner(cfg).run()` requiere que Othello-GPT esté cargado vía `TransformerLens`. Las celdas siguientes entrenan el SAE directamente sobre los archivos `.npy` pre-extraídos usando la arquitectura `BatchTopK` definida en el config.

```

NUM_EPOCHS = NAMING_META['extras']['epochs']

cfg = {
    'architecture':      'batchtopk',
    'd_in':              512,
    'd_sae':             4096,
    'k':                 50,
    'lr':                1e-3,
    'train_batch_size_tokens': 1024,
    'hook_name':         'blocks.5.hook_resid_post',
}

cfg_summary = pd.DataFrame({
    'Parámetro SAELens': ['architecture', 'd_in', 'd_sae', 'k (L0 promedio)', 'hook_name', 'lr'],
    'Valor': [cfg['architecture'], cfg['d_in'], cfg['d_sae'], cfg['k'],
              cfg['hook_name'], cfg['lr']]
})

print('Configuración SAELens:')
print(cfg_summary.to_string(index=False))

```

Configuración SAELens:

Parámetro SAELens	Valor
architecture	batchtopk
d_in	512
d_sae	4096
k (L0 promedio)	50
hook_name	blocks.5.hook_resid_post
lr	0.001

4. Dataset de Activaciones (Train/Validation Split 80-20%)

Activaciones pre-extraídas con `sae/activations/run_extraction.py` desde la capa `blocks.5.hook_resid_post`.

```

class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""
    def __init__(self, activations_path):
        self.activations = np.load(activations_path)
    def __len__(self):
        return len(self.activations)
    def __getitem__(self, idx):

```

```

        return torch.tensor(self.activations[idx], dtype=torch.float32)

activations_file = Path(f"../../activations/data/layer{NAMING_META['capa']}-{NAMING_META['id']}")
full_dataset = ActivationsDataset(activations_file)

# Split 80-20 reproducible
train_size = int(0.8 * len(full_dataset))
val_size    = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset, [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

train_loader = DataLoader(train_dataset, batch_size=cfg["train_batch_size_tokens"], shuffle=True)
val_loader   = DataLoader(val_dataset,   batch_size=cfg["train_batch_size_tokens"], shuffle=False)

dataset_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validación', 'Total'],
    'Muestras': [f'{len(train_dataset):,}', f'{len(val_dataset):,}', f'{len(full_dataset):,}'],
    'Shape':     [str(full_dataset.activations.shape), '', ''],
})
print(dataset_info.to_string(index=False))

```

```

Conjunto  Muestras      Shape
Train 4,720,000 (5900000, 512)
Validación 1,180,000
Total 5,900,000

```

5. Modelo: BatchTopK SAE

Implementación de la arquitectura BatchTopK de SAELens.

Diferencia clave vs ReLU + L1 y vs TopK estándar: - El encoder proyecta a `d_sae=8192` dimensiones (pre-activación) - La activación **BatchTopK** calcula un threshold global del batch (k-ésimo valor promediado); cada muestra puede tener más o menos de `k` features activas - `L0` es **variable por muestra**, pero el **promedio del batch** `k` (soft constraint a nivel de batch) - No hay `sparsity_coef` ni cosine annealing - El decoder se normaliza a norma unitaria después de cada paso del optimizador

```

class BatchTopKSAE(nn.Module):
    """

```

```

Sparse Autoencoder con activación BatchTopK.
Equivalente a SAE_Lens architecture='batchtopk'.

Encoder      : d_in -> d_sae (lineal + pre-bias centering)
Activación   : BatchTopK (mantiene el promedio de k features activas por batch)
Decoder      : d_sae -> d_in (columnas con norma unitaria)
"""
def __init__(self, d_in: int, d_sae: int, k: int, k_aux: int = 512):
    super().__init__()
    self.d_in = d_in
    self.d_sae = d_sae
    self.k = k
    self.k_aux = k_aux

    self.W_enc = nn.Parameter(torch.empty(d_in, d_sae))
    self.W_dec = nn.Parameter(torch.empty(d_sae, d_in))
    self.b_dec = nn.Parameter(torch.zeros(d_in))

    nn.init.kaiming_uniform_(self.W_enc)
    nn.init.kaiming_uniform_(self.W_dec)
    self.W_dec.data = self.W_dec.data / self.W_dec.data.norm(dim=-1, keepdim=True)
    self.register_buffer('num_batches_not_active', torch.zeros(d_sae))
    self.n_batches_to_dead = 10 # latente muerto si inactivo por 10 batches seguidos

def encode(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
    """Proyecta a espacio latente con activación BatchTopK"""
    x_cent = x - self.b_dec
    pre_acts = F.relu(x_cent @ self.W_enc)
    return self._topk_activation(pre_acts, threshold=threshold)

def _topk_activation(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
    """BatchTopK: selecciona exactamente k * batch_size activaciones usando scatter."""
    if threshold is None:
        acts_topk = torch.topk(x.flatten(), self.k * x.shape[0], dim=-1)
        return (
            torch.zeros_like(x.flatten())
            .scatter(-1, acts_topk.indices, acts_topk.values)
            .reshape(x.shape)
        )
    return torch.where(x > threshold, x, torch.zeros_like(x))

def decode(self, hidden: torch.Tensor) -> torch.Tensor:

```

```

        return hidden @ self.W_dec + self.b_dec

def preprocess_input(self, x: torch.Tensor):
    x_mean = x.mean(dim=-1, keepdim=True)
    x = x - x_mean
    x_std = x.std(dim=-1, keepdim=True)
    x = x / (x_std + 1e-5)
    return x, x_mean, x_std

def postprocess_output(self, x_reconstruct: torch.Tensor,
                       x_mean: torch.Tensor, x_std: torch.Tensor) -> torch.Tensor:
    return x_reconstruct * x_std + x_mean

def forward(self, x: torch.Tensor, threshold=None):
    x_norm, x_mean, x_std = self.preprocess_input(x)
    x_cent = x_norm - self.b_dec
    pre_acts = F.relu(x_cent @ self.W_enc)
    hidden = self._topk_activation(pre_acts, threshold=threshold)
    recon = self.postprocess_output(self.decode(hidden), x_mean, x_std)
    return recon, hidden, pre_acts

def auxiliary_loss(self, x: torch.Tensor, x_reconstruct: torch.Tensor, acts: torch.Tensor):
    """L_aux del repo oficial: reconstruye el residual con top-k_aux dead latents."""
    dead_features = self.num_batches_not_active >= self.n_batches_to_dead
    if dead_features.sum() == 0:
        return torch.tensor(0.0, device=x.device)
    residual = x.float() - x_reconstruct.float()
    acts_topk_aux = torch.topk(
        acts[:, dead_features],
        min(self.k_aux, int(dead_features.sum().item()))),
        dim=-1,
    )
    acts_aux = torch.zeros_like(acts[:, dead_features]).scatter(
        -1, acts_topk_aux.indices, acts_topk_aux.values
    )
    x_reconstruct_aux = acts_aux @ self.W_dec[dead_features]
    return (1/32) * (x_reconstruct_aux.float() - residual.float()).pow(2).mean()

@torch.no_grad()
def update_inactive_features(self, acts):
    self.num_batches_not_active += (acts.sum(0) == 0).float()
    self.num_batches_not_active[acts.sum(0) > 0] = 0

```

```

@torch.no_grad()
def estimate_threshold(self, loader, device, n_batches: int = 10) -> float:
    """Estima un threshold estable promediando sobre múltiples batches
    (independiente del batch actual - para usar en validación/inferencia)."""
    self.eval()
    thresholds = []
    for i, batch in enumerate(loader):
        if i >= n_batches:
            break
        x = batch.to(device)
        x_norm, _, _ = self.preprocess_input(x)
        x_cent = x_norm - self.b_dec
        pre_acts = F.relu(x_cent @ self.W_enc)
        mins = []
        for sample_idx in range(pre_acts.shape[0]):
            pos = pre_acts[sample_idx][pre_acts[sample_idx] > 0]
            if pos.numel() > 0:
                mins.append(pos.min().item())
        t = float(np.mean(mins)) if mins else 0.0
        thresholds.append(t)
    return float(np.mean(thresholds))

@torch.no_grad()
def make_decoder_weights_and_grad_unit_norm(self):
    """Proyecta el gradiente del decoder y renormaliza - igual que el repo oficial."""
    W_dec_normed = self.W_dec / self.W_dec.norm(dim=-1, keepdim=True)
    W_dec_grad_proj = (self.W_dec.grad * W_dec_normed).sum(-1, keepdim=True) * W_dec_normed
    self.W_dec.grad -= W_dec_grad_proj
    self.W_dec.data = W_dec_normed

# Crear modelo desde parámetros del cfg SAELens
model = BatchTopKSAE(d_in=cfg['d_in'], d_sae=cfg['d_sae'], k=cfg['k']).to(device)
num_params = sum(p.numel() for p in model.parameters())

model_info = pd.DataFrame({
    'Componente': [
        'Input', 'Encoder (W_enc)', 'Activación', 'Hidden activos', 'Decoder (W_dec)', 'Output'
    ],
    'Dimensión': [
        f'{cfg["d_in"]}',

```



```

        f'{cfg["d_in"]} -> {cfg["d_sae"]}',
        f'BatchTopK (k={cfg["k"]})',
        f'{cfg["d_sae"]} (solo {cfg["k"]} activos en promedio)',
        f'{cfg["d_sae"]} -> {cfg["d_in"]}',
        f'{cfg["d_in"]}',
        f'{model.k_aux}',
        f'{num_params:,'}
    ]
})
print(model_info.to_string(index=False))

```

Componente	Dimensión
Input	512
Encoder (W_enc)	512 -> 4096
Activación	BatchTopK (k=50)
Hidden activos	4096 (solo 50 activos en promedio)
Decoder (W_dec)	4096 -> 512
Output	512
k_aux	512
Total params	4,194,816

6. Función de Pérdida BatchTopK

Loss = MSE(reconstrucción)

- **MSE**: error de reconstrucción (igual que en L1)
- **Sin L1 penalty**: la sparsity ya está controlada por el threshold global de BatchTopK
- No hay `sparsity_coef` ni annealing — el único hiperparámetro de sparsity es `k` (promedio del batch)

```

ALPHA_AUX = 1 / 32 # peso de la pérdida auxiliar (arXiv:2412.06410)

def loss_function(x, reconstruction, acts, model):
    """
    Pérdida BatchTopK: MSE + alpha * L_aux (arXiv:2412.06410).
    - MSE      : error de reconstrucción
    - L_aux    : reconstruye el residual con las top-k_aux features para
                  evitar features muertas (alpha = 1/32)

    Args:
        x: Activaciones originales
    """

```

```

        reconstruction: Activaciones reconstruidas
        acts:           Pre-activaciones del encoder
        model:          Instancia de BatchTopKSAE
Returns:
    total_loss (scalar)
"""
mse    = F.mse_loss(reconstruction, x)
l_aux  = model.auxiliary_loss(x, reconstruction, acts)
return mse + l_aux

optimizer = torch.optim.Adam(model.parameters(), lr=cfg["lr"], betas=(0.9, 0.99))

optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning rate', 'betas', 'k (L0 promedio k)', 'k_aux', 'alpha_aux'],
    'Valor':      ['Adam', cfg["lr"], '(0.9, 0.99)', cfg["k"], model.k_aux, f'{ALPHA_AUX:.5f}']
})
print(optim_info.to_string(index=False))

```

	Parámetro	Valor
	Optimizador	Adam
	Learning rate	0.001
	betas	(0.9, 0.99)
k (L0 promedio k)		50
	k_aux	512
	alpha_aux	0.03125 (1/32)

7. Loop de Entrenamiento

```

PATIENCE = NAMING_META['extras']['patience']

save_dir = Path('./saved_model')
save_dir.mkdir(exist_ok=True)
best_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='best',
    extras_id=extras_id
)

```

```

)
BEST_CKPT = save_dir / best_model_name

history = {
    'train_mse_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_mse_loss': [],
    'val_l0_sparsity': [],
    'val_fraction_active': [],
}

best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

print(f'Iniciando entrenamiento: {NUM_EPOCHS} épocas máximo')
print(f'Early stopping: patience={PATIENCE}')
print(f'k = {cfg["k"]} features activas (L0 promedio k por batch)')
print(f'k_aux = {model.k_aux} | alpha_aux = {ALPHA_AUX:.5f} (1/32)')
print(f'Best checkpoint: {BEST_CKPT}')
print('=' * 70)

for epoch in range(NUM_EPOCHS):

    # TRAIN
    model.train()
    t_mse = t_l0 = t_frac = 0.0

    for batch in train_loader:
        x = batch.to(device)
        optimizer.zero_grad()
        recon, hidden, pre_acts = model(x)
        total_loss = loss_function(x, recon, pre_acts, model)
        total_loss.backward()
        model.make_decoder_weights_and_grad_unit_norm()
        optimizer.step()
        model.update_inactive_features(hidden)

    with torch.no_grad():
        t_mse += F.mse_loss(recon, x).item()
        t_l0 += (hidden > 0).float().sum(dim=1).mean().item()

```

```

        t_frac += (hidden > 0).float().mean().item()

n = len(train_loader)
history['train_mse_loss'].append(t_mse / n)
history['train_l0_sparsity'].append(t_l0 / n)
history['train_fraction_active'].append(t_frac / n)

# VALIDATION (sin threshold - igual que training, L0 k)
model.eval()
v_mse = v_l0 = v_frac = 0.0

with torch.no_grad():
    for batch in val_loader:
        x = batch.to(device)
        recon, hidden, _ = model(x)
        v_mse += F.mse_loss(recon, x).item()
        v_l0 += (hidden > 0).float().sum(dim=1).mean().item()
        v_frac += (hidden > 0).float().mean().item()

nv = len(val_loader)
history['val_mse_loss'].append(v_mse / nv)
history['val_l0_sparsity'].append(v_l0 / nv)
history['val_fraction_active'].append(v_frac / nv)

print(f"Época {epoch+1:3d}/{NUM_EPOCHS} | "
      f"Train MSE: {history['train_mse_loss'][-1]:.6f} | "
      f"Val MSE: {history['val_mse_loss'][-1]:.6f} | "
      f"Train L0: {history['train_l0_sparsity'][-1]:.1f} | "
      f"Val L0: {history['val_l0_sparsity'][-1]:.1f}")

# Early stopping
if history['val_mse_loss'][-1] < best_val_mse:
    best_val_mse = history['val_mse_loss'][-1]
    best_epoch = epoch + 1
    epochs_no_improve = 0

torch.save(model.state_dict(), BEST_CKPT)
save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],

```

```

        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best',
        extras=NAMING_META['extras'],
        config=cfg,
        history=history,
        optimizer=optimizer
    )
    print(f' Mejor modelo guardado (época {best_epoch} | Val MSE: {best_val_mse:.6f})')
else:
    epochs_no_improve += 1
    if epochs_no_improve >= PATIENCE:
        print(f'\nEarly stopping en época {epoch+1} (mejor: época {best_epoch})')
        break

# Restaurar mejor modelo
model.load_state_dict(torch.load(BEST_CKPT, weights_only=True))
print(f'\nMejor época: {best_epoch} | Mejor Val MSE: {best_val_mse:.6f}')

# Estimar threshold de inferencia al final del entrenamiento
inference_threshold = model.estimate_threshold(train_loader, device, n_batches=10)
print(f'Threshold de inferencia estimado: {inference_threshold:.4f}')

```

Iniciando entrenamiento: 1000 épocas máximo

Early stopping: patience=20

k = 50 features activas (L0 promedio k por batch)

k_aux = 512 | alpha_aux = 0.03125 (1/32)

Best checkpoint: saved_model\sae_topk_batch-topk_l6_100000g_ex1_best.pt

=====

Época 1/1000 | Train MSE: 0.191670 | Val MSE: 0.169610 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 1 | Val MSE: 0.169610)

Época 2/1000 | Train MSE: 0.166919 | Val MSE: 0.165411 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 2 | Val MSE: 0.165411)

Época 3/1000 | Train MSE: 0.164221 | Val MSE: 0.163811 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 3 | Val MSE: 0.163811)

Época 4/1000 | Train MSE: 0.163010 | Val MSE: 0.162963 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 4 | Val MSE: 0.162963)

Época 5/1000 | Train MSE: 0.162297 | Val MSE: 0.162398 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 5 | Val MSE: 0.162398)

Época 6/1000 | Train MSE: 0.161841 | Val MSE: 0.162067 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 6 | Val MSE: 0.162067)

Época 7/1000 | Train MSE: 0.161511 | Val MSE: 0.161796 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 7 | Val MSE: 0.161796)

Época 8/1000 | Train MSE: 0.161275 | Val MSE: 0.161757 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 8 | Val MSE: 0.161757)

Época 9/1000 | Train MSE: 0.161092 | Val MSE: 0.161488 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 9 | Val MSE: 0.161488)

Época 10/1000 | Train MSE: 0.160940 | Val MSE: 0.161436 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 10 | Val MSE: 0.161436)

Época 11/1000 | Train MSE: 0.160810 | Val MSE: 0.161190 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 11 | Val MSE: 0.161190)

Época 12/1000 | Train MSE: 0.160706 | Val MSE: 0.161256 | Train L0: 50.0 | Val L0: 50.0

Época 13/1000 | Train MSE: 0.160607 | Val MSE: 0.161075 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 13 | Val MSE: 0.161075)

Época 14/1000 | Train MSE: 0.160525 | Val MSE: 0.161004 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 14 | Val MSE: 0.161004)

Época 15/1000 | Train MSE: 0.160448 | Val MSE: 0.160931 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 15 | Val MSE: 0.160931)
 Época 16/1000 | Train MSE: 0.160392 | Val MSE: 0.160823 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 16 | Val MSE: 0.160823)
 Época 17/1000 | Train MSE: 0.160344 | Val MSE: 0.160839 | Train L0: 50.0 | Val L0: 50.0
 Época 18/1000 | Train MSE: 0.160303 | Val MSE: 0.160817 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 18 | Val MSE: 0.160817)
 Época 19/1000 | Train MSE: 0.160268 | Val MSE: 0.160790 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 19 | Val MSE: 0.160790)
 Época 20/1000 | Train MSE: 0.160227 | Val MSE: 0.160707 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 20 | Val MSE: 0.160707)
 Época 21/1000 | Train MSE: 0.160194 | Val MSE: 0.160832 | Train L0: 50.0 | Val L0: 50.0
 Época 22/1000 | Train MSE: 0.160161 | Val MSE: 0.160687 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 22 | Val MSE: 0.160687)
 Época 23/1000 | Train MSE: 0.160135 | Val MSE: 0.160813 | Train L0: 50.0 | Val L0: 50.0
 Época 24/1000 | Train MSE: 0.160106 | Val MSE: 0.160669 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 24 | Val MSE: 0.160669)
 Época 25/1000 | Train MSE: 0.160084 | Val MSE: 0.160640 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 25 | Val MSE: 0.160640)
 Época 26/1000 | Train MSE: 0.160061 | Val MSE: 0.160591 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 26 | Val MSE: 0.160591)
 Época 27/1000 | Train MSE: 0.160035 | Val MSE: 0.160582 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 27 | Val MSE: 0.160582)
 Época 28/1000 | Train MSE: 0.160017 | Val MSE: 0.160571 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt

Mejor modelo guardado (época 28 | Val MSE: 0.160571)
 Época 29/1000 | Train MSE: 0.159996 | Val MSE: 0.160539 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 29 | Val MSE: 0.160539)
 Época 30/1000 | Train MSE: 0.159986 | Val MSE: 0.160490 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 30 | Val MSE: 0.160490)
 Época 31/1000 | Train MSE: 0.159968 | Val MSE: 0.160507 | Train L0: 50.0 | Val L0: 50.0
 Época 32/1000 | Train MSE: 0.159957 | Val MSE: 0.160559 | Train L0: 50.0 | Val L0: 50.0
 Época 33/1000 | Train MSE: 0.159946 | Val MSE: 0.160407 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 33 | Val MSE: 0.160407)
 Época 34/1000 | Train MSE: 0.159932 | Val MSE: 0.160524 | Train L0: 50.0 | Val L0: 50.0
 Época 35/1000 | Train MSE: 0.159917 | Val MSE: 0.160497 | Train L0: 50.0 | Val L0: 50.0
 Época 36/1000 | Train MSE: 0.159907 | Val MSE: 0.160430 | Train L0: 50.0 | Val L0: 50.0
 Época 37/1000 | Train MSE: 0.159905 | Val MSE: 0.160417 | Train L0: 50.0 | Val L0: 50.0
 Época 38/1000 | Train MSE: 0.159888 | Val MSE: 0.160491 | Train L0: 50.0 | Val L0: 50.0
 Época 39/1000 | Train MSE: 0.159882 | Val MSE: 0.160513 | Train L0: 50.0 | Val L0: 50.0
 Época 40/1000 | Train MSE: 0.159875 | Val MSE: 0.160391 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 40 | Val MSE: 0.160391)
 Época 41/1000 | Train MSE: 0.159863 | Val MSE: 0.160376 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 41 | Val MSE: 0.160376)
 Época 42/1000 | Train MSE: 0.159861 | Val MSE: 0.160469 | Train L0: 50.0 | Val L0: 50.0
 Época 43/1000 | Train MSE: 0.159849 | Val MSE: 0.160373 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 43 | Val MSE: 0.160373)
 Época 44/1000 | Train MSE: 0.159840 | Val MSE: 0.160417 | Train L0: 50.0 | Val L0: 50.0
 Época 45/1000 | Train MSE: 0.159837 | Val MSE: 0.160436 | Train L0: 50.0 | Val L0: 50.0
 Época 46/1000 | Train MSE: 0.159828 | Val MSE: 0.160341 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 46 | Val MSE: 0.160341)
 Época 47/1000 | Train MSE: 0.159821 | Val MSE: 0.160432 | Train L0: 50.0 | Val L0: 50.0
 Época 48/1000 | Train MSE: 0.159815 | Val MSE: 0.160462 | Train L0: 50.0 | Val L0: 50.0
 Época 49/1000 | Train MSE: 0.159806 | Val MSE: 0.160405 | Train L0: 50.0 | Val L0: 50.0

Época 50/1000 | Train MSE: 0.159803 | Val MSE: 0.160390 | Train L0: 50.0 | Val L0: 50.0
 Época 51/1000 | Train MSE: 0.159799 | Val MSE: 0.160342 | Train L0: 50.0 | Val L0: 50.0
 Época 52/1000 | Train MSE: 0.159785 | Val MSE: 0.160357 | Train L0: 50.0 | Val L0: 50.0
 Época 53/1000 | Train MSE: 0.159781 | Val MSE: 0.160342 | Train L0: 50.0 | Val L0: 50.0
 Época 54/1000 | Train MSE: 0.159773 | Val MSE: 0.160476 | Train L0: 50.0 | Val L0: 50.0
 Época 55/1000 | Train MSE: 0.159772 | Val MSE: 0.160381 | Train L0: 50.0 | Val L0: 50.0
 Época 56/1000 | Train MSE: 0.159765 | Val MSE: 0.160420 | Train L0: 50.0 | Val L0: 50.0
 Época 57/1000 | Train MSE: 0.159754 | Val MSE: 0.160302 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 57 | Val MSE: 0.160302)
 Época 58/1000 | Train MSE: 0.159753 | Val MSE: 0.160325 | Train L0: 50.0 | Val L0: 50.0
 Época 59/1000 | Train MSE: 0.159744 | Val MSE: 0.160327 | Train L0: 50.0 | Val L0: 50.0
 Época 60/1000 | Train MSE: 0.159741 | Val MSE: 0.160331 | Train L0: 50.0 | Val L0: 50.0
 Época 61/1000 | Train MSE: 0.159732 | Val MSE: 0.160326 | Train L0: 50.0 | Val L0: 50.0
 Época 62/1000 | Train MSE: 0.159728 | Val MSE: 0.160347 | Train L0: 50.0 | Val L0: 50.0
 Época 63/1000 | Train MSE: 0.159728 | Val MSE: 0.160343 | Train L0: 50.0 | Val L0: 50.0
 Época 64/1000 | Train MSE: 0.159725 | Val MSE: 0.160337 | Train L0: 50.0 | Val L0: 50.0
 Época 65/1000 | Train MSE: 0.159721 | Val MSE: 0.160288 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 65 | Val MSE: 0.160288)
 Época 66/1000 | Train MSE: 0.159714 | Val MSE: 0.160311 | Train L0: 50.0 | Val L0: 50.0
 Época 67/1000 | Train MSE: 0.159716 | Val MSE: 0.160321 | Train L0: 50.0 | Val L0: 50.0
 Época 68/1000 | Train MSE: 0.159714 | Val MSE: 0.160336 | Train L0: 50.0 | Val L0: 50.0
 Época 69/1000 | Train MSE: 0.159705 | Val MSE: 0.160272 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 69 | Val MSE: 0.160272)
 Época 70/1000 | Train MSE: 0.159702 | Val MSE: 0.160322 | Train L0: 50.0 | Val L0: 50.0
 Época 71/1000 | Train MSE: 0.159701 | Val MSE: 0.160338 | Train L0: 50.0 | Val L0: 50.0
 Época 72/1000 | Train MSE: 0.159699 | Val MSE: 0.160373 | Train L0: 50.0 | Val L0: 50.0
 Época 73/1000 | Train MSE: 0.159689 | Val MSE: 0.160278 | Train L0: 50.0 | Val L0: 50.0
 Época 74/1000 | Train MSE: 0.159694 | Val MSE: 0.160306 | Train L0: 50.0 | Val L0: 50.0
 Época 75/1000 | Train MSE: 0.159695 | Val MSE: 0.160323 | Train L0: 50.0 | Val L0: 50.0
 Época 76/1000 | Train MSE: 0.159695 | Val MSE: 0.160310 | Train L0: 50.0 | Val L0: 50.0
 Época 77/1000 | Train MSE: 0.159690 | Val MSE: 0.160391 | Train L0: 50.0 | Val L0: 50.0
 Época 78/1000 | Train MSE: 0.159696 | Val MSE: 0.160311 | Train L0: 50.0 | Val L0: 50.0
 Época 79/1000 | Train MSE: 0.159689 | Val MSE: 0.160324 | Train L0: 50.0 | Val L0: 50.0
 Época 80/1000 | Train MSE: 0.159687 | Val MSE: 0.160263 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 80 | Val MSE: 0.160263)

Época 81/1000 | Train MSE: 0.159682 | Val MSE: 0.160293 | Train L0: 50.0 | Val L0: 50.0
 Época 82/1000 | Train MSE: 0.159685 | Val MSE: 0.160188 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 82 | Val MSE: 0.160188)
 Época 83/1000 | Train MSE: 0.159680 | Val MSE: 0.160288 | Train L0: 50.0 | Val L0: 50.0
 Época 84/1000 | Train MSE: 0.159679 | Val MSE: 0.160225 | Train L0: 50.0 | Val L0: 50.0
 Época 85/1000 | Train MSE: 0.159678 | Val MSE: 0.160330 | Train L0: 50.0 | Val L0: 50.0
 Época 86/1000 | Train MSE: 0.159673 | Val MSE: 0.160284 | Train L0: 50.0 | Val L0: 50.0
 Época 87/1000 | Train MSE: 0.159678 | Val MSE: 0.160302 | Train L0: 50.0 | Val L0: 50.0
 Época 88/1000 | Train MSE: 0.159675 | Val MSE: 0.160246 | Train L0: 50.0 | Val L0: 50.0
 Época 89/1000 | Train MSE: 0.159672 | Val MSE: 0.160306 | Train L0: 50.0 | Val L0: 50.0
 Época 90/1000 | Train MSE: 0.159668 | Val MSE: 0.160236 | Train L0: 50.0 | Val L0: 50.0
 Época 91/1000 | Train MSE: 0.159662 | Val MSE: 0.160288 | Train L0: 50.0 | Val L0: 50.0
 Época 92/1000 | Train MSE: 0.159662 | Val MSE: 0.160196 | Train L0: 50.0 | Val L0: 50.0
 Época 93/1000 | Train MSE: 0.159659 | Val MSE: 0.160234 | Train L0: 50.0 | Val L0: 50.0
 Época 94/1000 | Train MSE: 0.159660 | Val MSE: 0.160267 | Train L0: 50.0 | Val L0: 50.0
 Época 95/1000 | Train MSE: 0.159661 | Val MSE: 0.160238 | Train L0: 50.0 | Val L0: 50.0
 Época 96/1000 | Train MSE: 0.159657 | Val MSE: 0.160281 | Train L0: 50.0 | Val L0: 50.0
 Época 97/1000 | Train MSE: 0.159658 | Val MSE: 0.160183 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 97 | Val MSE: 0.160183)
 Época 98/1000 | Train MSE: 0.159658 | Val MSE: 0.160253 | Train L0: 50.0 | Val L0: 50.0
 Época 99/1000 | Train MSE: 0.159653 | Val MSE: 0.160227 | Train L0: 50.0 | Val L0: 50.0
 Época 100/1000 | Train MSE: 0.159652 | Val MSE: 0.160376 | Train L0: 50.0 | Val L0: 50.0
 Época 101/1000 | Train MSE: 0.159654 | Val MSE: 0.160167 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\100000games\sae_topk_batch-topk_l6_100000g_ex1_best.pt
 Mejor modelo guardado (época 101 | Val MSE: 0.160167)
 Época 102/1000 | Train MSE: 0.159651 | Val MSE: 0.160322 | Train L0: 50.0 | Val L0: 50.0
 Época 103/1000 | Train MSE: 0.159648 | Val MSE: 0.160231 | Train L0: 50.0 | Val L0: 50.0
 Época 104/1000 | Train MSE: 0.159645 | Val MSE: 0.160293 | Train L0: 50.0 | Val L0: 50.0
 Época 105/1000 | Train MSE: 0.159645 | Val MSE: 0.160256 | Train L0: 50.0 | Val L0: 50.0
 Época 106/1000 | Train MSE: 0.159647 | Val MSE: 0.160226 | Train L0: 50.0 | Val L0: 50.0
 Época 107/1000 | Train MSE: 0.159644 | Val MSE: 0.160309 | Train L0: 50.0 | Val L0: 50.0
 Época 108/1000 | Train MSE: 0.159651 | Val MSE: 0.160287 | Train L0: 50.0 | Val L0: 50.0
 Época 109/1000 | Train MSE: 0.159642 | Val MSE: 0.160235 | Train L0: 50.0 | Val L0: 50.0
 Época 110/1000 | Train MSE: 0.159642 | Val MSE: 0.160197 | Train L0: 50.0 | Val L0: 50.0
 Época 111/1000 | Train MSE: 0.159645 | Val MSE: 0.160308 | Train L0: 50.0 | Val L0: 50.0
 Época 112/1000 | Train MSE: 0.159636 | Val MSE: 0.160220 | Train L0: 50.0 | Val L0: 50.0
 Época 113/1000 | Train MSE: 0.159642 | Val MSE: 0.160222 | Train L0: 50.0 | Val L0: 50.0
 Época 114/1000 | Train MSE: 0.159639 | Val MSE: 0.160314 | Train L0: 50.0 | Val L0: 50.0

Época 115/1000 | Train MSE: 0.159631 | Val MSE: 0.160308 | Train L0: 50.0 | Val L0: 50.0
Época 116/1000 | Train MSE: 0.159631 | Val MSE: 0.160236 | Train L0: 50.0 | Val L0: 50.0
Época 117/1000 | Train MSE: 0.159632 | Val MSE: 0.160221 | Train L0: 50.0 | Val L0: 50.0
Época 118/1000 | Train MSE: 0.159629 | Val MSE: 0.160251 | Train L0: 50.0 | Val L0: 50.0
Época 119/1000 | Train MSE: 0.159624 | Val MSE: 0.160234 | Train L0: 50.0 | Val L0: 50.0
Época 120/1000 | Train MSE: 0.159623 | Val MSE: 0.160186 | Train L0: 50.0 | Val L0: 50.0
Época 121/1000 | Train MSE: 0.159618 | Val MSE: 0.160178 | Train L0: 50.0 | Val L0: 50.0

Early stopping en época 121 (mejor: época 101)

Mejor época: 101 | Mejor Val MSE: 0.160167

Threshold de inferencia estimado: 0.0024

8. Guardar Modelo

```
# Guardar checkpoint final
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': {
        'architecture': cfg["architecture"],
        'd_in': cfg["d_in"],
        'd_sae': cfg["d_sae"],
        'k': cfg["k"],
        'lr': cfg["lr"],
        'hook_name': cfg["hook_name"],
    },
    'history': history,
}

final_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras_id=extras_id
)

checkpoint_path = save_dir / final_model_name
torch.save(checkpoint, checkpoint_path)
print(f'Modelo guardado: {checkpoint_path}')
```

```

save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras=NAMING_META['extras'],
    config=cfg,
    history=history,
    optimizer=optimizer
)

```

Modelo guardado: saved_model\sae_topk_batch-topk_l6_100000g_ex1_final.pt

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_batch-topk_l6_100000games\sae_topk_batch-topk_l6_100000g_ex1_final.pt

WindowsPath('C:/Users/Esposa/Documents/Repos/sae-othello-gpt/layer_06/models/sae_topk/sae_topk_batch-topk_l6_100000games/sae_topk_batch-topk_l6_100000g_ex1_final.pt')

9. Métricas básicas del modelo

```

import json

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

training_metrics = {
    # Calidad de reconstrucción
    'val_mse_final': history['val_mse_loss'][-1],
    'val_mse_best': best_val_mse,
    'val_l0_final': history['val_l0_sparsity'][-1],
}

```

```

'val_fraction_active_final': history['val_fraction_active'][-1],
# Contexto del entrenamiento
'best_epoch': best_epoch,
'total_epochs': len(history['train_mse_loss']),
'train_mse_final': history['train_mse_loss'][-1],
'train_l0_final': history['train_l0_sparsity'][-1],
# Hiperparámetros TopK
'architecture': cfg["architecture"],
'k': cfg["k"],
'd_sae': cfg["d_sae"],
'd_in': cfg["d_in"],
'expansion_factor': cfg["d_sae"] // cfg["d_in"],
'hook_name': cfg["hook_name"],
}

os.makedirs(metrics_dir, exist_ok=True)
metrics_path = Path(metrics_dir) / 'training_metrics.json'
with open(metrics_path, 'w') as f:
    json.dump(training_metrics, f, indent=2)

print(f'Métricas guardadas: {metrics_path}')
print(pd.DataFrame(
    list(training_metrics.items()), columns=['Métrica', 'Valor']
).to_string(index=False))

```

Métricas guardadas: c:\Users\Esposa\Documents\Repos\othello_world\sae\models\experiments\lay
topk_topk\100000games\ex1\metrics\training_metrics.json

Métrica	Valor
val_mse_final	0.160178
val_mse_best	0.160167
val_l0_final	50.0
val_fraction_active_final	0.012207
best_epoch	101
total_epochs	121
train_mse_final	0.159618
train_l0_final	50.0
architecture	batchtopk
k	50
d_sae	4096
d_in	512
expansion_factor	8
hook_name	blocks.5.hook_resid_post

10. Visualización de Resultados

```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

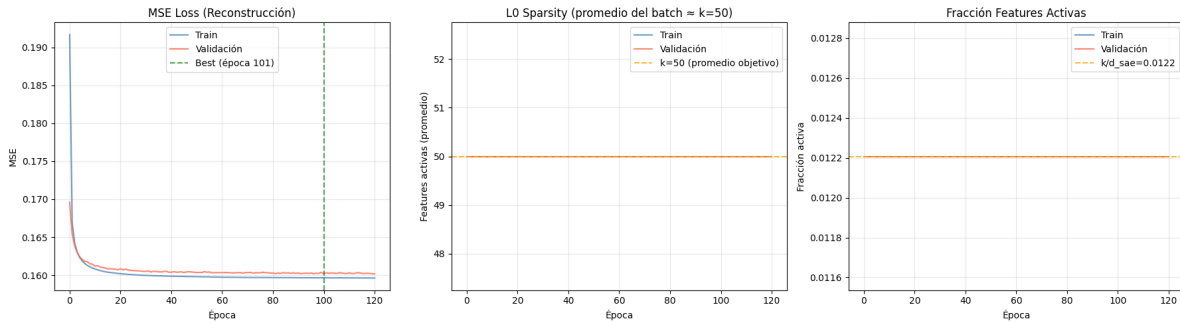
# MSE Loss
axes[0].plot(history['train_mse_loss'], label='Train', color='steelblue', alpha=0.8)
axes[0].plot(history['val_mse_loss'], label='Validación', color='tomato', alpha=0.8)
if best_epoch > 0:
    axes[0].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7,
                    label=f'Best (época {best_epoch})')
axes[0].set_title('MSE Loss (Reconstrucción)')
axes[0].set_xlabel('Época')
axes[0].set_ylabel('MSE')
axes[0].legend()
axes[0].grid(True, alpha=0.3)

# L0 sparsity (para BatchTopK el promedio del batch k, pero varía por muestra)
axes[1].plot(history['train_l0_sparsity'], label='Train', color='steelblue', alpha=0.8)
axes[1].plot(history['val_l0_sparsity'], label='Validación', color='tomato', alpha=0.8)
axes[1].axhline(cfg["k"], color='orange', linestyle='--', alpha=0.8, label=f'k={cfg["k"]} (p')
axes[1].set_title(f'L0 Sparsity (promedio del batch k={cfg["k"]})')
axes[1].set_xlabel('Época')
axes[1].set_ylabel('Features activas (promedio)')
axes[1].legend()
axes[1].grid(True, alpha=0.3)

# Fracción activa
axes[2].plot(history['train_fraction_active'], label='Train', color='steelblue', alpha=0.8)
axes[2].plot(history['val_fraction_active'], label='Validación', color='tomato', alpha=0.8)
expected_frac = cfg["k"] / cfg["d_sae"]
axes[2].axhline(expected_frac, color='orange', linestyle='--', alpha=0.8,
                label=f'k/d_sae={expected_frac:.4f}')
axes[2].set_title('Fracción Features Activas')
axes[2].set_xlabel('Época')
axes[2].set_ylabel('Fracción activa')
axes[2].legend()
axes[2].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(plots_dir / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()
```

```
print('\n' + '='*50)
print('Gráficas guardadas exitosamente')
print('='*50)
```



```
=====
Gráficas guardadas exitosamente
=====
```

11. Resumen Final

```
print('\n' + '='*70)
print(' RESUMEN FINAL DEL ENTRENAMIENTO (BatchTopK SAE)')
print('='*70)

arch_summary = pd.DataFrame({
    'Componente': ['Input (d_in)', 'Hidden (d_sae)', 'k (L0 promedio k)', 'Expansión', 'Par
    'Valor': [
        f'{cfg["d_in"]}',
        f'{cfg["d_sae"]}',
        f'{cfg["k"]} features activas (promedio del batch)',
        f'{cfg["d_sae"] // cfg["d_in"]}x',
        f'{num_params:,}'
    ]
})
print('\nArquitectura:')
print(arch_summary.to_string(index=False))

data_summary = pd.DataFrame({
    'Conjunto': ['Train', 'Validación', 'Total'],
```

```

    'Muestras': [f'{len(train_dataset):,}', f'{len(val_dataset):,}', f'{len(full_dataset):,}']
})
print('\nDatos:')
print(data_summary.to_string(index=False))

final_summary = pd.DataFrame({
    'Métrica': [
        'Val MSE best', 'Val MSE final',
        'Val L0 final (promedio k)', 'Val Fracción activa', 'Mejor época'
    ],
    'Valor': [
        f'{best_val_mse:.6f}',
        f'{history['val_mse_loss'][-1]:.6f}',
        f'{history['val_l0_sparsity'][-1]:.1f} (k={cfg['k']})',
        f'{history['val_fraction_active'][-1]:.4%}',
        f'{best_epoch}/{len(history['train_mse_loss'])}'
    ]
})
print('\nMétricas:')
print(final_summary.to_string(index=False))

```

```

=====
RESUMEN FINAL DEL ENTRENAMIENTO (BatchTopK SAE)
=====

```

Arquitectura:

Componente	Valor
Input (d_in)	512
Hidden (d_sae)	4096
k (L0 promedio k) 50 features activas (promedio del batch)	
Expansión	8x
Parámetros	4,194,816

Datos:

Conjunto	Muestras
Train	4,720,000
Validación	1,180,000
Total	5,900,000

Métricas:

Métrica	Valor
---------	-------

Val MSE best	0.160167
Val MSE final	0.160178
Val L0 final (promedio k) 50.0 (k=50)	
Val Fracción activa	1.2207%
Mejor época	101/121

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```
import subprocess

exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_batch-topk_v7.ipynb'
output_path = exp_dir / pdf_name

print(f'Generando PDF con Quarto...')
print(f'Archivo de salida: {output_path}')

result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
```

```
)  
  
if result.returncode == 0:  
    print(f'PDF generado exitosamente: {output_path}')  
else:  
    print(f'Error al generar PDF:')  
    print(result.stderr)
```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\models\experiments\layer
topk_topk\100000games\ex1\sae_topk_batch-topk_l6_100000g_ex1_training.pdf
PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\models\experiments
topk_topk\100000games\ex1\sae_topk_batch-topk_l6_100000g_ex1_training.pdf